

CM0246 Discrete Structures

Cardinality

Andrés Sicard-Ramírez

Universidad EAFIT

Semestre 2014-2

Preliminaries

Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Cardinality

Definition

Let A be a finite set. The number of (distinct) elements in A , denoted $|A|$, is called the **cardinality** of A .

Cardinality

Definition

Let A be a finite set. The number of (distinct) elements in A , denoted $|A|$, is called the **cardinality** of A .

Definition

Let A and B be finite or infinite sets. The sets A and B have the **same cardinality**, if and only, there is a **bijection** from A to B .

Cardinality

Definition

Let A be a finite set. The number of (distinct) elements in A , denoted $|A|$, is called the **cardinality** of A .

Definition

Let A and B be finite or infinite sets. The sets A and B have the **same cardinality**, if and only, there is a **bijection** from A to B .

Injunction, surjection or bijection?

Draw figures in the whiteboard.

Cardinality

Example

(Proofs on the whiteboard)

- $|\mathbb{Z}^+| = |\mathbb{N}|$.

Cardinality

Example

(Proofs on the whiteboard)

- $|\mathbb{Z}^+| = |\mathbb{N}|$.
- $|\mathbb{N}| = |\text{Even}|$, where $\text{Even} = \{ 2n \mid n \in \mathbb{N} \}$.

Example

(Proofs on the whiteboard)

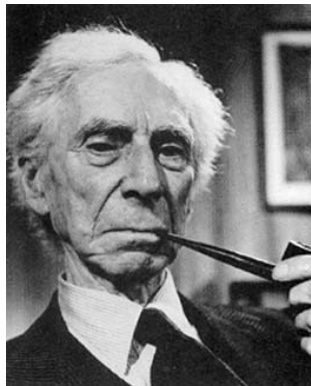
- $|\mathbb{Z}^+| = |\mathbb{N}|$.
- $|\mathbb{N}| = |\text{Even}|$, where $\text{Even} = \{ 2n \mid n \in \mathbb{N} \}$.
- $|\mathbb{N}| = |M_k|$, where M_k is the set of the non-negative multiples of $k \in \mathbb{Z}^+$, i.e., $M_k = \{ nk \mid n \in \mathbb{N} \}$.

Cardinality

Example

(Proofs on the whiteboard)

- $|\mathbb{Z}^+| = |\mathbb{N}|$.
- $|\mathbb{N}| = |\text{Even}|$, where $\text{Even} = \{ 2n \mid n \in \mathbb{N} \}$.
- $|\mathbb{N}| = |M_k|$, where M_k is the set of the non-negative multiples of $k \in \mathbb{Z}^+$, i.e., $M_k = \{ nk \mid n \in \mathbb{N} \}$.
- $|[0, 1]| = |[a, b]|$, where $a, b \in \mathbb{R}$ and $a < b$.



(1872 – 1970)[†]

‘The possibility that whole and part may have the same number of terms is, it must be confessed, shocking to common-sense.’ (Russell 1903, p. 358)

[†]Image from the MacTutor History of Mathematics Archive.

Cardinality

Example (Lipschutz ([1964] 1998), Solved problem 6.2, p. 153)

Show that $|[0, 1]| = |(0, 1)|$.

Cardinality

Example (Lipschutz ([1964] 1998), Solved problem 6.2, p. 153)

Show that $|[0, 1]| = |(0, 1)|$.

Solution

Note that

$$\begin{aligned}[0, 1] &= \{0, 1, 1/2, 1/3, 1/4, \dots\} \cup A \\ (0, 1) &= \{1/2, 1/3, 1/4, \dots\} \cup A\end{aligned}$$

where

$$\begin{aligned}A &= [0, 1] - \{0, 1, 1/2, 1/3, 1/4, \dots\} \\ &= (0, 1) - \{1/2, 1/3, 1/4, \dots\}.\end{aligned}$$

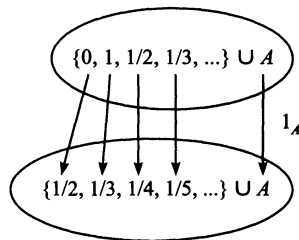
(continued on next slide)

Cardinality

Solution (continuation)

From the figure[†] we define the bijective function $f : [0, 1] \rightarrow (0, 1)$ by

$$f(x) = \begin{cases} 1/2, & \text{if } x = 0; \\ 1/(n+1), & \text{if } x = 1/n \text{ where } n \in \mathbb{Z}^+; \\ x, & \text{otherwise.} \end{cases}$$



[†]Figure source: (Lipschutz [1964] 1998, Fig. 6.5).

Cardinality

Exercise

Let A and B be sets. Show $|A \times B| = |B \times A|$.

Enumerable and Non-Enumerable Sets

Question

Has all the infinite sets the same cardinality?

Enumerable and Non-Enumerable Sets

Question

Has all the infinite sets the same cardinality?

Definition

A set that is either **finite** or has **the same cardinality as the set of positive integers** is called **enumerable** (or **countable**).

Enumerable and Non-Enumerable Sets

Question

Has all the infinite sets the same cardinality?

Definition

A set that is either **finite** or has **the same cardinality as the set of positive integers** is called **enumerable** (or **countable**).

Definition

A set that is not enumerable (not countable) is called **non-enumerable** (or **uncountable**).

Enumerable and Non-Enumerable Sets

Question

Has all the infinite sets the same cardinality?

Definition

A set that is either **finite** or has **the same cardinality as the set of positive integers** is called **enumerable** (or **countable**).

Definition

A set that is not enumerable (not countable) is called **non-enumerable** (or **uncountable**).

Example

Whiteboard.

Enumerable and Non-Enumerable Sets

Theorem

The interval $(0, 1)$ is non-enumerable.

(continued on next slide)

Enumerable and Non-Enumerable Sets

Proof.

Let's suppose $(0, 1)$ is enumerable.

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14}\dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24}\dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34}\dots$$

$$\vdots$$

Enumerable and Non-Enumerable Sets

Proof.

Let's suppose $(0, 1)$ is enumerable.

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14} \dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24} \dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34} \dots$$

$$\vdots$$

Let $r = 0.d_1d_2d_3 \dots \in (0, 1)$, where

$$d_i = \begin{cases} 4, & \text{if } d_{ii} \neq 4; \\ 5, & \text{if } d_{ii} = 4. \end{cases}$$

Enumerable and Non-Enumerable Sets

Proof.

Let's suppose $(0, 1)$ is enumerable.

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14} \dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24} \dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34} \dots$$

$$\vdots$$

Let $r = 0.d_1d_2d_3 \dots \in (0, 1)$, where

$$d_i = \begin{cases} 4, & \text{if } d_{ii} \neq 4; \\ 5, & \text{if } d_{ii} = 4. \end{cases}$$

The number r does not belong to the above enumeration. Therefore $(0, 1)$ is non-enumerable. 

Enumerable and Non-Enumerable Sets

Theorem

Let A and B be sets such that $A \subseteq B$. If A is non-enumerable then B is non-enumerable.

Enumerable and Non-Enumerable Sets

Theorem

The set of the real numbers is non-enumerable.

Enumerable and Non-Enumerable Sets

Theorem

The set of the real numbers is non-enumerable.

Proof.

The interval $(0, 1)$ is a non-enumerable subset of $\mathbb{R}$. Therefore (using a previous theorem), $\mathbb{R}$ is non-enumerable. ■

Enumerable and Non-Enumerable Sets

Theorem

The set of the real numbers is non-enumerable.

Proof.

The interval $(0, 1)$ is a non-enumerable subset of $\mathbb{R}$. Therefore (using a previous theorem), $\mathbb{R}$ is non-enumerable. ■

Remark

Comment about the continuum hypothesis.

Enumerable and Non-Enumerable Sets

Remark

The quadratic formulae are the solution to the quadratic equation

$$ax^2 + bx + c = 0.$$

Two quadratic formulae are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

and

$$x = \frac{-2c}{b \mp \sqrt{b^2 - 4ac}}. \tag{1}$$

Enumerable and Non-Enumerable Sets

Example

Show that $|(-1, 1)| = |\mathbb{R}|$.

Enumerable and Non-Enumerable Sets

Example

Show that $|(-1, 1)| = |\mathbb{R}|$.

Solution

The function $f : (-1, 1) \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{x}{1 - x^2}$$

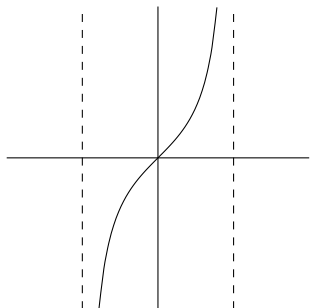
has as inverse the function $f^{-1} : \mathbb{R} \rightarrow (-1, 1)$ given by (obtained using the quadratic formula (1))

$$f^{-1}(x) = \frac{2x}{1 + \sqrt{1 + 4x^2}}.$$

(continued on next slide)

Enumerable and Non-Enumerable Sets

Solution (continuation)



$$f(x) = \frac{x}{1-x^2}$$

Since the function f is a bijection then $|(-1, 1)| = |\mathbb{R}|$. Source: Munkres ([1974] 2000, Example § 18.5).

References

- Seymour Lipschutz [1964] (1998). Set Theory and Related Topics. 2nd ed. Schaum's Outline. McGraw-Hill (cit. on pp. 11–13).
- James R. Munkres [1974] (2000). Topology. 2nd ed. Prentice Hall (cit. on p. 31).
- Kenneth H. Rosen [1988] (2004). Matemática Discreta y sus Aplicaciones. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).
- [1988] (2012). Discrete Mathematics and Its Applications. 7th ed. McGraw-Hill (cit. on p. 19).
- Bertrand Russell (1903). The Principles of Mathematics. Cambridge University Press (cit. on p. 10).